Test-time physics-informed fine-tuning of neural PDE surrogates under a misspecified coefficient

Nikolaus Wehofer[1], *Anna Zimmel*[2]

[1] Johannes Kepler Universität Linz (jku.at)

[2] Johannes Kepler Universität Linz, Institute of Bioinformatics (bioinf.jku.at)

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Keywords: test-time adaptation, physics-informed fine-tuning, neural PDE surrogates, physics residual, model misspecification, distribution shift, Kolmogorov flow, predictability horizon

TL;DR: A Kolmogorov-flow surrogate fine-tuned at five times its training Reynolds number on a misspecified residual, from five unlabeled initial conditions, extends predictability at every resolved scale. The true viscosity is never used.

Abstract:

Neural PDE surrogates degrade outside their training regime, where shifts often arrive unlabeled and the changed physical coefficient is unknown. Although the governing equation provides a label-free training signal, a changed coefficient makes its residual misspecified. We show that minimizing this residual can still help. Using limited, unlabeled initial conditions, we fine-tune a pretrained Kolmogorov-flow surrogate at five times its training Reynolds number. Even with the original viscosity in the residual, adaptation extends predictability at every resolved spectral scale. The gain requires over 200 steps, far more than one-shot test-time adaptation methods. At smaller budgets, the largest scales briefly regress, with no warning from the residual. Contrary to the general assumption, using the correct target viscosity yields no further benefit. Thus, when the shifted term contributes little to the residual magnitude, adaptation may require neither detecting the shift nor estimating the changed coefficient.

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Paper Decision

Decision by Program Chairs 29 Sep 2026 at 18:45 (modified: 29 Sep 2026 at 22:47) Program Chairs, Reviewers, Authors Revisions

Decision: Reject

Comment:

Reject: The reviews were mixed, and on balance the paper is not yet ready for the workshop.

The demonstrated effect is plausible, but the experiments don't identify the mechanisms

Official Review by Reviewer om7y 22 Sep 2026 at 14:03 (modified: 29 Sep 2026 at 22:45) Program Chairs, Reviewers, Authors Revisions

Summary:

The paper adapts a Re=100 Kolmogorov surrogate to Re=500 by descending a PDE residual, plus an IC anchor, over five unlabeled target initial states, keeping the source viscosity. Per-shell correlation horizons lengthen, and using the true target viscosity gives no gain on this readout. The issue of of the paper is not the numbers but whether the protocol isolates the claimed phenomenon.

Review:

The overall investigation is careful and diagnostic. The authors show the correlation/amplitude split, the threshold-sensitivity check, and the IC-pool and residual-viscosity controls make the effect easy to inspect, and the limitations paragraph pre-empts many objections. But the following three confounds jointly undercut the strongest reading.

1. Physics vs. regularizer is under-determined by the paper's own ablation. If no diffusion term case still recovers 35% of the headroom and an over-diffusive case (further from the target than the source) is the best tested value at 50%, the residual's viscosity is acting as a tunable smoothness knob whose optimum is unrelated to the true physics. The paper rejects this via an "interior optimum," but a well-tuned regularizer is exactly an interior optimum, so that rebuttal doesn't discriminate the hypotheses.
2. The headline is evaluated with a metric that is deliberately amplitude-invariant. Ek in Eq5 only reads the phase term, while viscosity controls dissipation and thus spectral amplitudes.
3. Operator shift is entangled with an input-distribution shift. Each regime is tested for 100 time units using its own viscosity before the initial states are collected. This means the five target initial vorticity fields already come from a different state distribution. Since using source-regime initial conditions makes performance worse, it is unclear whether the method is adapting to the changed dynamics or simply using differences already present in the initial vorticity fields.

Strengths And Weaknesses:

Strengths

1. Good workshop fit
2. Great transparency. The correlation/amplitude decomposition and the viscosity and source-IC controls probe the mechanism rather than only reporting an aggregate
3. The found negative and coefficient-agnostic result is interesting and discussion-provoking

Weaknesses

1. The central interpretation ("physics despite misspecification") is not separated from three competitors, namely, a smoothness regularizer near-optimally tuned at the source kinematic viscosity, adaptation to the target initial vorticity field marginal, and a phase-only readout that hides the physical effect of viscosity. Each of them is individually plausible, but together they mean the causal reading isn't yet earned
2. The "recovered fraction of headroom" is signed, uncapped, and summed over a hand-chosen band (k2–k8), while the whole-field readout is null/negative exactly at the budgets where the method can hurt. This is a researcher-degrees-of-freedom concern
3. Unfortunately, most load-bearing numbers live in appendix tables

Quality: 3: good
Clarity: 2: fair
Significance: 3: good
Originality: 3: good
Relevance: 4: Relevant
Limitations:
Yes

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.
Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
Ethical Concerns: No ethical concerns
Code Of Conduct: I acknowledge and agree
Responsible Reviewing: I acknowledge and agree

A case study of adaptation with a misspecified residual; mechanism, utility, and generality remain open.

Official Review by Reviewer x6ay 21 Sep 2026 at 18:03 (modified: 29 Sep 2026 at 22:45) Program Chairs, Reviewers, Authors Revisions

Summary:

This paper investigates physics-informed test-time adaptation of neural PDE surrogates under a shift in viscosity. The authors fine-tune a 3D U-Net pretrained on two-dimensional Kolmogorov flow at $Re = 100$, using five initial-state snapshots from independent $Re = 500$ simulations, each collected after 100 time units of burn-in. Adaptation uses an initial-condition loss and a PDE residual that retains the source viscosity, without using target rollout trajectories in the gradient updates.

The main empirical finding is that this procedure improves scale-resolved correlation horizons on held-out target trajectories and, under the tested optimization protocol, yields greater improvements than using the correct target viscosity. The work explores a potential route to reusing pretrained surrogates when target-state observations are available but target rollout trajectories are scarce.

Review:

General Evaluation

The paper presents an interesting case study on adapting neural PDE surrogates when the physical regime shifts, target trajectories are unavailable, and governing parameters are unknown. The premise fits the workshop's focus on imperfect simulators by testing whether an intentionally misspecified residual can guide adaptation from limited initial-state observations. The finding that retaining the source viscosity outperforms the target viscosity raises a provocative question. However, the study does not establish the underlying physical mechanism, leaves flow energetics and amplitude accuracy unverified, and relies on deployment assumptions that require clarification.

Pros

- **Held-out target evaluation:** Adaptation is evaluated on 30 held-out target trajectories, confirming benefits extend beyond the five adaptation snapshots.
- **Scale-resolved analysis:** Measuring correlation horizons across Fourier shells reveals nuanced scale-dependent dynamics and early large-scale degradation obscured by whole-field metrics.
- **Informative ablations:** The authors provide useful ablations on viscosity values, loss components, and pool composition, alongside candid acknowledgment of limitations.

Cons

- **Incompletely assessed physical accuracy:** Longer correlation horizons do not establish accurate flow amplitudes or energetics. While the authors cite Lippe et al. [7] to justify correlation horizons over L_2 error, Lippe et al. crucially paired correlation tracking with spatial Fourier energy spectra. Without held-out relative L_2 errors or energy/enstrophy diagnostics, it remains unclear whether gains reflect genuine physical adaptation or implicit Laplacian damping from the higher source viscosity.
- **Unclear practical motivation and deployment premise:** The premise lacks a concrete physical application where full-field snapshots are available without temporal rollouts or known parameters. The paper does not address why a practitioner would fine-tune with a misspecified residual rather than estimating the coefficient directly, nor does it account for the 100 burn-in solver units required to generate target states. Target-supervised learning rate selection also qualifies the label-free claim.
- **Limited empirical scope:** Evidence is restricted to a single PDE, architecture, and $Re = 100 \to 500$ shift. While internal ablations are informative, reverse-direction transfer ($Re = 500 \to 100$) or additional target regimes are needed to determine whether this behavior generalizes beyond this specific setup.

Suggestions for Authors

1. **Ground the deployment scenario:** Introduce a concrete physical application where full-field snapshots exist without trajectories or known parameters, address snapshot generation costs, and evaluate a pre-fixed or source-selected learning rate.
2. **Report physical accuracy diagnostics:** Complement correlation horizons with held-out relative L_2 errors and energy/enstrophy spectra to verify amplitude preservation, and use shared color limits across methods in Figure 3.
3. **Isolate the adaptation mechanism:** Run a reverse-direction shift ($Re = 500 \to 100$) and sweep the loss-weight ratio ($\lambda_{\mathrm{ic}}/\lambda_{\mathrm{pde}}$) to test whether source-viscosity gains stem from implicit regularization.

Strengths And Weaknesses:

Strengths

- **Quality:** Held-out evaluation on 30 target trajectories, paired test-chain uncertainty estimates, and viscosity, loss-component, and pool ablations support the reported correlation gains.
- **Clarity:** The setup, adaptation objective, and scale-resolved evaluation are clearly described, with explicit acknowledgment of early degradation and deployment limitations.
- **Significance:** The observed benefit of a misspecified residual is relevant to the workshop and motivates further study of surrogate reuse with imperfect physical models.
- **Originality:** The source-versus-target viscosity comparison and its scale-dependent outcomes offer an interesting empirical perspective on physics-informed adaptation.

Weaknesses

- **Quality:** Evidence is limited to one PDE and one transfer case. Although alternate-pool tests are provided, it remains unclear how reliably five target snapshots support adaptation to the changed dynamics, rather than improved prediction from target-like states. The gains may partly reflect regularization from the higher source viscosity; a reverse-transfer experiment would help assess this possibility. Correlation horizons demonstrate improved structural agreement, but amplitude-sensitive errors and energy diagnostics are needed to establish broader physical accuracy.
- **Clarity:** No major concerns. The paper is generally clear, and the main reservations concern the experimental evidence and practical interpretation.
- **Significance:** The single transfer setting and lack of a concrete deployment example leave broader usefulness uncertain. Practical value also depends on how target snapshots are acquired and the associated costs.
- **Originality:** No major concerns. The adaptation objective build on existing physics-informed fine-tuning; but using the misspecified coefficient covers the originality.

Quality: 2: fair
Clarity: 3: good
Significance: 3: good
Originality: 3: good
Relevance: 5: Highly relevant
Limitations:
yes

Rating: 4: Borderline accept: Technically solid paper where reasons to accept outweigh reasons to reject, e.g., limited evaluation. Please use sparingly.
Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
Ethical Concerns: No ethical concerns
Code Of Conduct: I acknowledge and agree
Responsible Reviewing: I acknowledge and agree

A good workshop fit and intresting results, but evidence is limited to a single PDE/architecture/shift setting

Official Review by Reviewer rWec 17 Sep 2026 at 19:39 (modified: 29 Sep 2026 at 22:45) Program Chairs, Reviewers, Authors Revisions

Summary:

The central result of the paper is interesting and well aligned with workshop. On Kolmogorov flow shifted from Re=100 to Re=500, fine-tuning on only five unlabelled target initial conditions using the source-regime viscosity improves prediction horizons across the reported spectral scales, while surprisingly using the correct target viscosity provides no additional benefit

Review:

The experimental methodology is strong. The evaluation uses 30 held-out target trajectories and reports paired bootstrap intervals over 4000 resamples. The spectral-shell correlation horizons are also appropriate choice for chaotic dynamics. The ablations are particularly useful.

Cons:

-- code does not set a seed and that every reported cell corresponds to one unseeded run, the 3-repeat check is not enough

-- main results rely on a particularly favorable kind of misspecification

-- no comparison to the external TTA baselines it positions against

-- the originality is somewhat limited

Strengths And Weaknesses:

Strengths

-- good thematic fit to the workshop

-- reproducibility is comparatively strong, there are useful ablations

-- the central empirical result is non-obvious

Weaknesses

-- only one PDE family, one architecture, one shift. Generalization is unclear.

-- there is no observable stopping criterion, the method requires a budget specified in advance, there is no answer which checkpoint has the best predictive skill

-- missing baselines, at least one external baseline on the same Kolmogorov setup would strengthen the case

Quality: 3: good
Clarity: 2: fair
Significance: 3: good
Originality: 2: fair
Relevance: 4: Relevant
Limitations:
yes

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.
Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
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